

Lab 3

Create a Custom VPC and Deploy EC2

AWS Technical Essentials • Lab 3 of 6

🏷️ Resource Names for This Lab

Replace `<ID>` with your assigned student ID (for example `user7`).

Resource	Name
VPC wizard — name tag auto-generation	`employee-directory-<ID>`
Resulting VPC	`employee-directory-<ID>-vpc`
Security group (new, in your VPC)	`employee-directory-vpc-sg-<ID>`
EC2 instance (Name tag)	`employee-directory-app-vpc-<ID>`

> 💡 **Nice side effect:** the wizard applies your name tag to every resource it creates — subnets, route tables and the internet gateway all pick up your ID automatically.

📋 Lab Overview

🎯 Objectives

- Create a VPC with public subnets using the wizard
- Enable auto-assign public IPv4 on both public subnets
- Create a security group in the new VPC
- Launch an EC2 instance in the custom VPC
- Verify the web application works

📝 Lab Details

- **Duration:** ~25 minutes
- **Region:** `us-east-1` (N. Virginia) — same for every lab
- **Prerequisite:** Lab 2 instance running

> ⚠️ Keep this VPC — it is needed for Lab 6.

🔧 Create VPC — Settings

1. Navigate to **VPC** in the console
2. Click **Create VPC**
3. Select **"VPC and more"**

Setting	Value
Name tag auto-generation	`employee-directory-<ID>`
IPv4 CIDR block	`10.1.0.0/16`
Number of Availability Zones	**2**
Number of public subnets	**2**
Number of private subnets	**0**
NAT gateways	**None**

🔧 Create VPC — Execute

1. Review the **Preview** panel on the right
2. Click **Create VPC**
3. Wait for all resources to show **green checkmarks**
4. Click **View VPC**

> ✅ **Expected:** VPC created with 2 public subnets, an internet gateway, and route tables — all auto-configured.

🌐 Enable Auto-Assign Public IPv4

The wizard leaves this **off**. Turn it on now — Lab 6 launches instances from a template that inherits the subnet setting, and without a public IP they cannot reach the package repos.

1. In the VPC console left nav, click **Subnets**
2. Filter by `<ID>` and select your first public subnet
3. **Actions → Edit subnet settings**
4. Check **Enable auto-assign public IPv4 address → Save**
5. Repeat for your second public subnet

> ✅ **Expected:** both of your public subnets show **Auto-assign public IPv4: Yes**.

🔒 Create Security Group

1. In the VPC console left nav, click **Security groups**
2. Click **Create security group**
3. Name: `employee-directory-vpc-sg-<ID>` — a *new* group, not the Lab 2 one
4. Description: *Allow HTTP access for Employee Directory*
5. VPC: Select **employee-directory-<ID>-vpc**

Type	Port	Source
HTTP	80	0.0.0.0/0
SSH	22	My IP

6. Click **Create security group**

🚀 Launch EC2 in the New VPC

1. Navigate to **EC2** → **Launch instance**
2. Name: `employee-directory-app-vpc-<ID>`
3. AMI: **Amazon Linux 2023**
4. Instance type: **t2.micro**
5. Key pair: **employee-dir-key-<ID>** (from Lab 2)
6. Under **Network settings**, click **Edit**:
    - VPC: **employee-directory-<ID>-vpc**
    - Subnet: Choose any **public** subnet
    - Auto-assign public IP: **Enable**
    - Select existing security group: **employee-directory-vpc-sg-<ID>**

🚀 Add User Data and Launch

1. Under **Advanced details**, scroll to **User data** and paste:

```
#!/bin/bash
dnf install -y httpd
systemctl enable --now httpd
echo '<!DOCTYPE html>
<html><head><title>Employee Directory</title>
<style>body{font-family:Arial,sans-serif;max-width:800px;margin:50px auto;padding:20px}
h1{color:#1a365d}table{border-collapse:collapse;width:100%}th,td{border:1px solid #ddd;padding:12px;text-align:left}
th{background:#2b6cb0;color:white}tr:nth-child(even){background:#f7fafc}</style></head>
<body><h1>Employee Directory</h1>
<p>Welcome to the Employee Directory Application</p>
<table><tr><th>Name</th><th>Department</th><th>Location</th></tr>
<tr><td>John Smith</td><td>Engineering</td><td>Seattle</td></tr>
<tr><td>Jane Doe</td><td>Marketing</td><td>New York</td></tr>
<tr><td>Bob Johnson</td><td>Sales</td><td>Austin</td></tr></table>
</body></html>' > /var/www/html/index.html
```

3. Click **Launch instance**

🌐 Test the Web Server

1. Wait for the new instance to reach **Running** with **2/2 checks passed**
2. Copy the **Public IPv4 address**
3. Open a browser tab: `http://<PUBLIC_IP>`

> ✅ **Expected:** The Employee Directory page loads — now running inside your custom VPC.

> 🕒 **Note:** Allow 1-2 minutes after "Running" for Apache to finish installing.

🛑 Stop the Lab 2 Instance

1. In the EC2 console, find your original **employee-directory-app-<ID>** instance (from Lab 2)
2. Select it, click **Instance state** → **Stop instance**

> 💡 **Why?** The Lab 2 instance was in the default VPC. Now that you have a working instance in your custom VPC, you no longer need the old one running. Stopping saves cost.

⚠️ Resources to Keep

> **Do NOT delete these resources** — they are needed for Lab 6:

- **VPC:** employee-directory-<ID>-vpc (with subnets, IGW, route tables)
- **Security Group:** employee-directory-vpc-sg-<ID>
- **Key Pair:** employee-dir-key-<ID>

You may terminate the EC2 instance from this lab — Lab 6 will create new instances via Auto Scaling.

🎉 Lab 3 Complete!

Custom VPC with EC2 Deployed

🎯 Key Takeaways

🛠️ What You Did

- Created a VPC with the "VPC and more" wizard
- Configured 2 public subnets across 2 AZs
- Created a security group in the new VPC
- Launched and tested an EC2 instance

📚 Key Concepts

- **VPC** = isolated network in the cloud
- **Public subnets** have routes to an IGW
- **Multi-AZ** enables high availability
- **Security groups** are VPC-specific



Questions?

Ask your instructor before proceeding to the next module